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From: Siena Romano, Pennsylvania State University MAS Student Consultant

To: Mayor Daniel Laurie and the San Francisco Traffic Stop Analysis Committee

San Francisco Stop Trends, Forecasts, and Associations with Driver Demographics

EXECUTIVE SUMMARY

Purpose and Scope: In response to Mayor Daniel Laurie's Street Safety Initiative, the San Francisco Traffic Stop Analysis Committee and a Penn State MAS student conducted a comprehensive study of historical traffic stop data (January 2007 – June 2016) to establish a baseline when evaluating future stop trends, stop volumes, and equitable outcomes across different driver demographics.

Analysis of Monthly Seasonality: To identify patterns in stop frequency, a Negative Binomial Regression (NBR) model was utilized. This approach accounted for the over-dispersed nature of the data where the variance in monthly stops exceeded the mean.

- **Key Finding:** Across the ten-year study period, stop counts during June, November, and December were consistently lower than the rest of the year (with 90% confidence), suggesting a seasonal dip in enforcement or driver activity during the early summer and end-of-year holiday periods.

Forecasting Future Stop Volume: The study employed an ARIMA (Autoregressive Integrated Moving Average) model to provide a benchmark for future months. This model accounts for both short-term fluctuations and annual seasonal cycles.

- **Application:** Based on historical data, the model projected that for July 2016, the city should expect between 3,888 and 5,416 moving violations with 90% confidence.

Demographic Disparities and Citation Rates: The final phase of the study examined whether a driver's race and gender influenced the likelihood of a stop resulting in a formal citation.

- **Key Finding:** Three-way chi-square tests for independence and Cramer's V statistics concluded that there was a significant, moderately sized relationship between race and receiving a citation from a moving violation for male drivers. Specifically, pairwise comparisons revealed notable disparities in citation likelihood for **Asian/Pacific Islander vs. Black** drivers and **Black vs. "Other"** race drivers. These findings highlight specific areas where the city should monitor for potential bias as enforcement increases.

1.0 – PROJECT DESCRIPTION

The San Francisco Traffic Stop Analysis Committee is conducting ongoing testing over counts, trends, and recipients of moving violations due to SF Mayor Daniel Laurie's new Street Safety Initiative. The Initiative was enacted in December 2025 to increase street safety for all SF residents by coordinating city police departments and stop policies, implementing automated speed cameras, and improving streets and walkways[1]. As one of the directive's goals is to reduce traffic collisions, Laurie has increased police presence and cameras throughout the city, and he expects the number of traffic citations to rise in 2026.

Laurie believes that an increase in traffic stops can lead to disproportionate profiling and citations that are not distributed for public safety but to meet department demands. Therefore, he requested that the Committee, with the assistance of a Penn State MAS student, analyze SF traffic stop data from 10 to 20 years ago to confirm if stop disparity and spikes in stops are worse, the same, or better following the new initiative. Further, Laurie asked that a model be built on historical data to see how well it can predict an interval of the expected number of moving violations per month as he'll want to check if future stop counts are outside of the appropriate range. As the historical analyses being conducted were for exploratory research with low-risk policy impacts and cost of false positives, extreme statistical precision was not required, and 90% confidence was accepted for all statistical tests.

The study used January 2007 through June 2016 SF stop data provided by a team of researchers from the Stanford Open Policing Project who collected and standardized law enforcement stops across the United States for public large scale analyses[2]. While the full dataset included all traffic stops for moving and non-moving violations, only moving violations were analyzed as auto mechanical issues or traffic collisions are unrelated to increased policing. After the initial filtering, 563,315 records remained for testing if there was a trend in when stops occurred, if the count of moving violations could be predicted, and if there were relationships between driver race and gender and receiving a citation following a moving violation.

Laurie plans to begin comparative analyses six months after the start of the directive (May 2026) when enough recent traffic citation data is collected. As a result, the historical trend analysis, model building, and association testing must conclude in April 2026.

1.1 – RESEARCH QUESTIONS

1. Was there a monthly trend in counts of traffic stops for moving violations in San Francisco between 2007 and 2016?
2. Can the expected number of moving violations in San Francisco be forecasted with a prediction interval for a month outside of the data's time period, July 2016?
3. Were there three-way interactions between drivers' race and sex and receiving a traffic citation following a moving violation in San Francisco between 2007 and 2016?

1.2 – VARIABLES

The traffic stop data was sourced from the Stanford Open Policing Project[3] (OpenPoliceData · Streamlit) and included dates, locations, driver demographics, reason for stops, and outcomes of traffic stops that occurred in San Francisco between January 2007 and June 2016. After filtering the data for only moving violations, stop counts per month were calculated by aggregating the number of rows by year and month and log transformed to improve data normality. See Table 1.1 for the variables used in the analyses along with their descriptions, data types, and value ranges.

Table 1.1 Traffic Stop Variables, Descriptions, Types, and Range of Values

Variable	Description	Type	Range/Notes
Month Year	Month and year of stop	Quantitative – Discrete; Explanatory	Jan 2007 – Jun 2016
Stop Count	Number of stops	Quantitative – Discrete; Response	Natural Log; 8.122 – 8.865
Subject Race	Race of driver pulled over	Categorical – Nominal; Explanatory/Control	asian/pacific islander, black, hispanic, other, white
Subject Sex	Sex of driver pulled over	Categorical – Binary; Explanatory/Control	male, female
Citation Issued	Flag indicating if the driver was cited	Categorical – Binary; Response	TRUE, FALSE
Reason for Stop	Reason the driver was pulled over	Categorical – Nominal; Control	Value includes "moving violation"

2.0 – EXPLORATORY DATA ANALYSIS

The underlying data was explored prior to performing statistical analyses and modeling. Specifically, monthly stop counts were log transformed to reduce variance and summarized from January 2007 to June 2016 for research question 1 and from July 2011 to June 2016 for research question 2. Unlike research question 1 requesting evidence of monthly trends going back the full ten years, only five years of data were input into the time series models for research question 2 to improve forecasting accuracy with recent years and reduce the impact of outdated monthly patterns. Less years also prevented the models from overfitting to irrelevant noise.

Additionally, logged count outliers that were more than three standard deviations away from the mean were identified and removed (Sep 2013, 8.05 and Dec 2013, 8.00). Refer to Table 2.1 and Table 2.2 below for the min, max, mean, interquartile range, and standard deviation of the moving violation counts for the two time periods.

Table 2.1 Summary Statistics of Monthly Stop Counts January 2007 – June 2016

Scale	Min	1st Quartile	Median	Mean	3rd Quartile	Max	Std Dev.
Untransformed	3,367	4,507	4,996	4,912	5,503	7,081	786
Natural Log	8.122	8.413	8.516	8.499	8.613	8.865	0.161

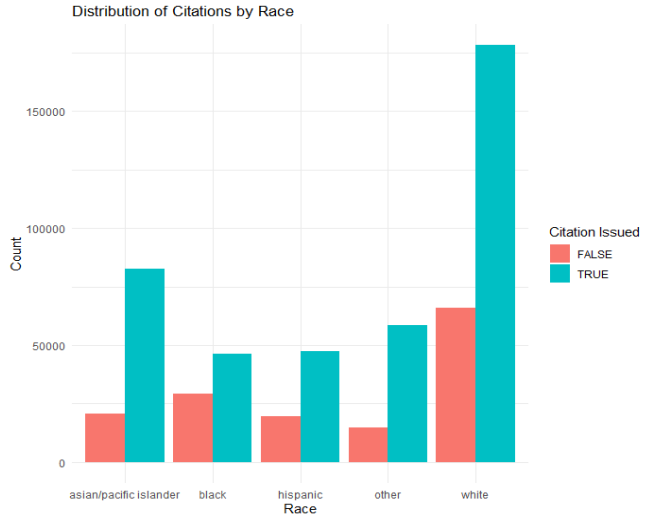
Table 2.2 Summary Statistics of Monthly Stop Counts July 2011 – June 2016

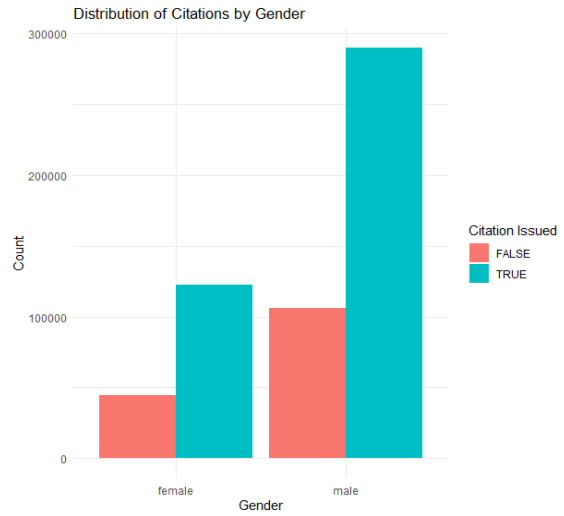
Scale	Min	1st Quartile	Median	Mean	3rd Quartile	Max	Std Dev.
Untransformed	2,989	3,949	4,500	4,445	4,951	6,015	669
Natural Log	8.003	8.281	8.412	8.388	8.507	8.702	0.154

Monthly traffic stop data was further evaluated to determine normality using histograms and boxplots. Refer to Figures 2.1 and 2.2 in the Appendix depicting a normal bell curve in the histograms of moving violations for the two time periods. Figures 2.3 and 2.4 in the Appendix also provide a visual representation of the range of stop counts and lack of outliers for the two time periods prior to statistical analyses. Last, the scatterplot in Figure 2.5 in the Appendix shows no visual trends that would impact modeling of monthly stops between July 2011 and June 2016.

To understand the breakdown of stop citations by driver demographic for research question 3, counts of drivers pulled over for moving violations were aggregated by gender and race and further grouped by whether a citation was given or not. Refer to Tables 2.3 and 2.4 in the Appendix for the citation counts by driver race and sex. Additionally, for a visual representation of the distribution of citations, Figures 2.6 and 2.7 below show comparative bar graphs by race and gender. Based on the distribution counts and graphs, the number of white and male drivers pulled over for moving violations were noticeably higher than the stop counts of all other drivers.

Figures 2.6 and 2.7 Comparative Bar Graphs of Citations by Race and Gender





3.0 – STATISTICAL ANALYSIS

3.1 – Was there a monthly trend in counts of traffic stops for moving violations in San Francisco between 2007 and 2016?

NBR with months as the predictor was performed to confirm if there were significant count differences in moving violations between months, regardless of the year, from January 2007 to June 2016. To test that the data passed NBR assumptions, no significant outliers were found using cook's distance as distances were under 0.5 (shown in Figure 3.1 in the Appendix). Also, a likelihood ratio test between the NBR model and a Poisson model with months as a predictor[4] confirmed that traffic stop counts were appropriately over dispersed and NBR was a better fit (chi-squared p-value of 0). Last, a heteroskedasticity-consistent covariance matrix estimator was added to the NBR model to handle autocorrelation and produce robust standard errors[5].

The NBR results shown in Table 3.1 indicate that months June, November, and December were significantly lower over the ten years. Specifically, at 90% confidence, the expected number of stops was about 15% less in June, 12% less in November, and 17% less in December in reference to the incidence rate ratio (IRR). None of the months had significantly higher stop trends. Refer to Table 3.2 in the Appendix for the NBR results for all months besides the reference group (Jan).

Table 3.1 Significant Negative Binomial Regression Results

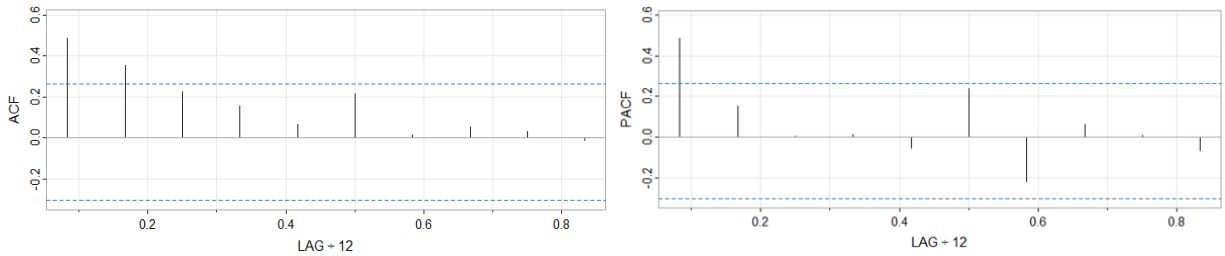
	Jun	Nov	Dec
Estimate	-0.161	-0.128	-0.185
IRR	0.851	0.880	0.831
Robust Std. Error	0.065	0.077	0.079
P-Value	0.013	0.095	0.019

To confirm that the trends identified were truly significant and not caused by overfitting, a likelihood ratio test was conducted between the NBR model with months as a predictor and an NBR null model (with intercept only)[4]. The results showed that the model with months as a predictor was significantly different than the null model with a chi-square p-value of 0.072. Therefore, at 90% confidence, the NBR results above were significant trends.

3.2 – Can the expected number of moving violations in San Francisco be forecasted with a prediction interval for a month outside of the data's time period, July 2016?

An ARIMA model with SF traffic stop data from July 2011 to June 2016 was used to forecast the interval of expected number of moving violations in July 2016. Specifically, an ARIMA model with non-season AR(2) and MA(1) terms, a seasonal AR(2) term, and a seasonal period of 12 had the lowest MAE (0.094), MAPE (0.011), and RMSE (0.125) in comparison to other ARIMA, GARCH, and triple exponential smoothing models. Additionally, the selected model's results passed the assumptions of no trend in the scatterplot, no significant autocorrelations in the ACF of residuals, normally distributed residuals in the QQ plot, and non-significant residuals in the Ljung-Box plot[6] shown and further discussed in Figure 3.2 in the Appendix. These plots, along with the ACF and PACF shown in Figure 3.3 depicting stationary data (white noise) with lags tailing to zero[6], confirm that the ARIMA model was appropriate for the data.

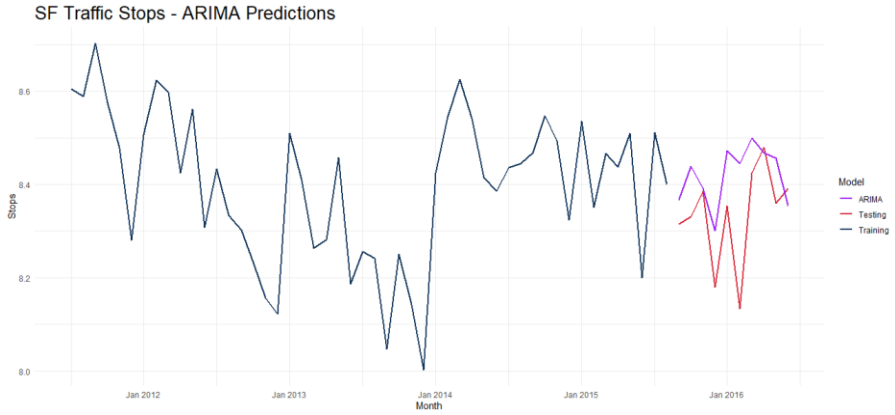
Figure 3.3 ACF and PACF



Before forecasting the prediction interval for July 2016, the ARIMA model was trained on moving violation data from July 2011 to August 2015 (85% of the stops). Figure 3.4 shows a line

plot of the ARIMA model's average forecast for each stop count compared to the true test set[7] (stop counts from September 2015 to June 2016).

Figure 3.4 Line Plot of ARIMA Average Forecasts in the Test Set



The ARIMA model forecasted 12 months of moving violations using the July 2011 to June 2016 data. Figure 3.5 in the Appendix shows those prediction intervals with one to two standard deviations from July 2016 to June 2017. After applying the exponential function to the natural log counts, the upper and lower bounds for July 2016 were calculated using a predicted value of 4,652 and a margin of error of 764 and confirmed at 90% confidence that the count of moving violations would be between 3,888 and 5,416.

3.3 – Were there three-way interactions between drivers' race and sex and receiving a traffic citation following a moving violation in San Francisco between 2007 and 2016?

To measure the association between driver sex, race, and receiving a traffic citation following a moving violation between January 2007 and June 2016, three-way chi-square tests for independence and Cramer's V statistics were calculated to determine if the variables were conditionally associated and not the same across driver demographics and to conclude the strength of those interactions. These tests were appropriate as the data passed key requirements of expected counts greater than five and observations were collected independently[8] with rows representing unique drivers pulled from all areas of the city by different police officers. The chi-square tests resulted in p-values of approximately 0 for all relationships, meaning there were significant differences in the associations between driver race and receiving a citation across male and female drivers and between driver gender and receiving a citation across all races.

The Cramer's V statistics were also analyzed because it determined the true effect size of the interactions and is necessary when performing chi-square tests that are highly sensitive to large sample sizes (563,315). The results in Table 3.3 show that effect size of the associations between driver gender and receiving a citation were very low across all races given a degree of freedom of 1 and results below 0.1, the low effect size threshold. This means that the relationship strengths were weak[9] and the associations between driver gender and receiving a citation across all races were likely independent and not changing across race levels.

Table 3.3 Association of Driver Gender and Receiving a Citation Given Race

	Asian/Pacific Islander	**Black**	**Hispanic**	**Other**	**White**
Cramer's V	0.0159	0.0339	0.0219	0.0170	0.0084

Further, the results in Table 3.4 show that the effect size of the associations between driver race and receiving a citation for males were medium with degrees of freedom of 4 and results around 0.15, the medium effect size threshold. This means that the magnitudes of the differences in associations for males were moderate[9], indicating that the associations between driver race and receiving a citation can change for male drivers. The effect size was low for females as the statistic was further under 0.15.

Table 3.4 Association of Driver Race and Receiving a Citation Given Gender

	Female	**Male**
Cramer's V	0.1123	0.1415

After identifying that there was a medium sized effect in the relationship between race and receiving a citation for males, pairwise comparisons between races and citations were performed to determine which associations between race and receiving a citation were significantly different. Based on the results, all races had significant differences in the associations between each other and receiving a citation with p-values of approximately 0. Using the Cramer's V statistics to measure the strength of those associative differences, Table 3.5 in the Appendix shows that the relationship between Asian/Pacific Islander and Black male drivers and receiving a citation and the relationship between Black and other race male drivers and receiving a citation had approximate medium effect sizes (statistics around 0.3). This means that the magnitudes of the differences in associations were moderate[9] and the associations between Asian/Pacific Islander and Black drivers and between Black and other race drivers and receiving a citation were different for male drivers during the time period.

4.0 – RECOMMENDATIONS

1. *Was there a monthly trend in counts of traffic stops for moving violations in San Francisco between 2007 and 2016?*

Based on the NBR results, the count of moving violations in June, November, and December were significantly lower than other months between January 2007 and June 2016. Specifically, June had about 15% less, November had about 12% less, and December had about 17% less stops, indicating negative trends for those months.

2. *Can the expected number of moving violations in San Francisco be forecasted with a prediction interval for a month outside of the data's time period, July 2016?*

Using monthly stop data between July 2011 to June 2016, an ARIMA model predicted that July 2016 would have between 3,888 and 5,416 moving violations with 90% confidence.

3. *Were there three-way interactions between drivers' race and sex and receiving a traffic citation following a moving violation in San Francisco between 2007 and 2016?*

Chi-squared tests for independence and Cramer's V statistics provided evidence that there was a moderately strong and significant three-way relationship between race and receiving a citation for male drivers. Further analyses using pairwise comparisons and Cramer's V statistics confirmed that there were significant differences in the associations between Asian/Pacific Islander and Black male drivers and receiving a citation and between Black and other raced male drivers and receiving a citation on a moderate level.

5.0 – RESOURCES

While completing statistical testing, modeling, and output interpretations for the report's research questions, the following academic books, papers, and R and SAS tutorials were referenced:

Negative Binomial Regression	• *Regression Analysis of Count Data* by A. Colin Cameron and Pravin K. Trivedi • "Negative Binomial Regression | R Data Analysis Examples" from UCLA's
ARIMA Model Forecasting	• *Forecasting: Principles and Practice* by Rob J Hyndman and George Athanasopoulos • "Introducing ARIMA Models" by Joshua Noble
Chi-Square Three-Way Test of Association and Cramer's V Statistic	• *Statistical Notes for Clinical Researchers: Chi-Squared Test and Fisher's Exact Test* by Hae-Young Kim • *An Introduction to Categorical Analysis* by Alan Agresti

6.0 – CONSIDERATIONS

6.1 – POLICY AND OPERATIONAL CONSIDERATIONS

SF law enforcement policy changes between January 2007 and June 2016 related to department quotas, traffic stops, or citations could be a lurking variable that impacted trend analysis or independence testing across the ten years. For example, there could have been a significant trend in when tickets were given out some months and not all because of changes in department objectives. Further, changes in policy around ticketing residents of different races and sexes could have impacted independence testing of these features and receiving a citation.

Additionally, the ARIMA model was not trained on all ten years of data due to the high variance in stop count across the ten years. A better predictive time series model could be produced with more data points if Laurie and the SF Traffic Stop Analysis Committee removed years with policies aimed at reducing traffic stops. This is because the years with initiatives to reduce stops do not represent the trend of increased traffic citations that are expected with the new directive.

6.2 – METHODOLOGICAL CONSIDERATIONS

Future analysts should note that there is an imbalance in counts between driver race and sex. While the Cramer's V statistic measures the effect size independently of total sample size, the test can be influenced by smaller or larger counts in groups and indicate a higher association. However, as the results did not show a very high effect size between any of the groups, there is no large concern that historical findings were inaccurate.

Further, while significant differences were identified in the interactions between Asian/Pacific Islander and Black male drivers and receiving a citation and between Black and other raced male drivers and receiving a citation on a medium sized level, the results do not tell us which race increases or decreases the chances of receiving a ticket. Specifically, the current study only presents the differences and magnitudes of the relationship between races and receiving a moving violation. Future research would be required to identify which races were most at risk of receiving a moving violation between January 2007 and June 2016.

7.0 – APPENDIX

7.1 – TABLES

Table 2.3 Observation Count of Driver Race When a Citation was Issued or Not

	Asian/Pacific Islander	Black	Hispanic	Other	White
Cited	82,557	46,130	47,427	58,416	178,236
Not Cited	20,823	29,352	19,625	14,716	66,033
Total	103,380	75,482	67,052	73,132	244,269

Table 2.4 Observation Count of Driver Gender When a Citation was Issued or Not

	Male	Female
Cited	290,155	122,611
Not Cited	106,009	44,540
Total	396,164	167,151

Table 3.2 Negative Binomial Regression Results

	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Estimate	-0.053	0.022	-0.056	0.008	-0.161	-0.082	-0.058	-0.028	-0.085	-0.128	-0.185
IRR	0.948	1.022	0.946	1.008	0.851	0.921	0.944	0.972	0.918	0.880	0.831
Robust Std. Error	0.069	0.068	0.060	0.070	0.065	0.059	0.076	0.080	0.073	0.077	0.079
P-Value	0.440	0.745	0.352	0.915	0.013	0.163	0.448	0.723	0.241	0.095	0.019

Table 3.5 Pairwise Comparison of Race and Receiving a Citation with Cramer's V Statistics

Paired Races	Cramer's V
Asian/Pacific Islander; Black	0.2222
Asian/Pacific Islander; Hispanic	0.1170
Asian/Pacific Islander; Other	0.0095
Asian/Pacific Islander; White	0.0753
Black; Hispanic	0.1062
Black; Other	0.2127
Black; White	0.1239
Hispanic; Other	0.1079
Hispanic; White	0.0291
Other; White	0.0638

7.2 – FIGURES

Figure 2.1 Histogram of Natural Log Stop Counts by Month January 2007 – June 2016

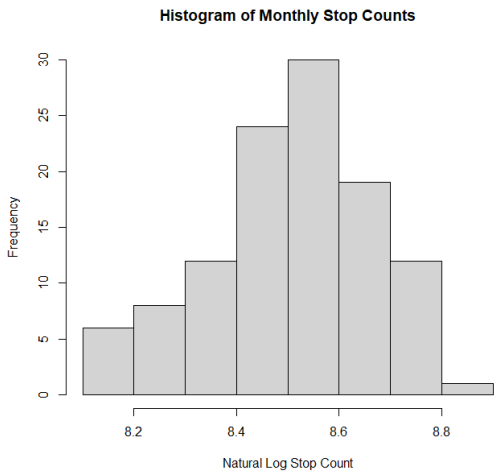


Figure 2.2 Histogram of Natural Log Stop Counts by Month for July 2011 – June 2016

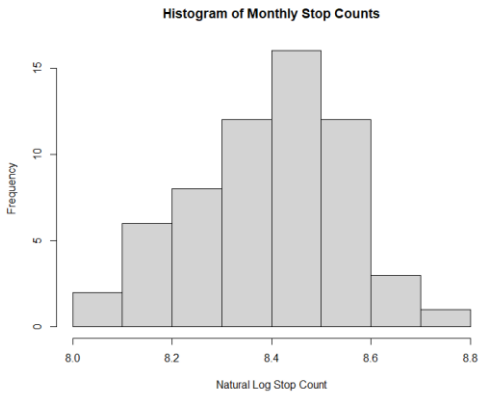


Figure 2.3 Boxplot of Natural Log Stop Counts by Month January 2007 – June 2016

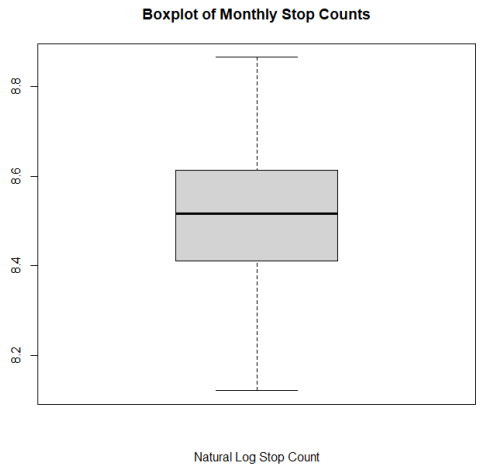


Figure 2.4 Boxplot of Natural Log Stop Counts by Month for July 2011 – June 2016

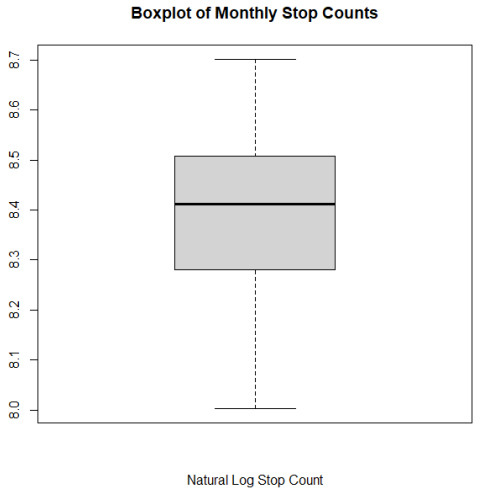


Figure 2.5 Scatterplot of Natural Log Stop Counts by Month for July 2011 – June 2016

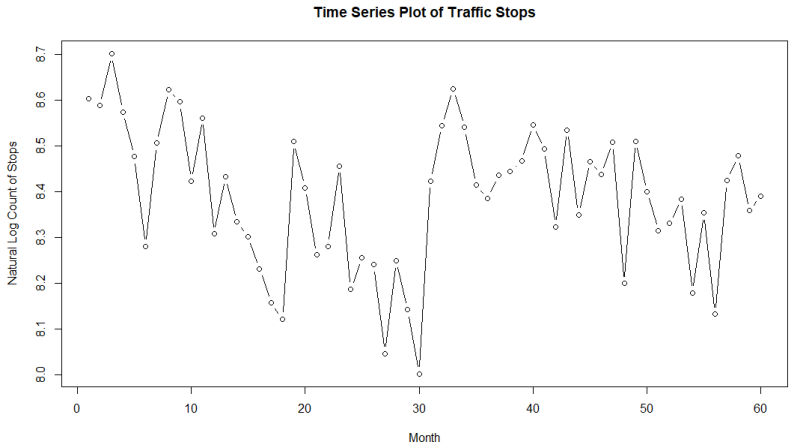


Figure 3.1 Cook's Distance Plot

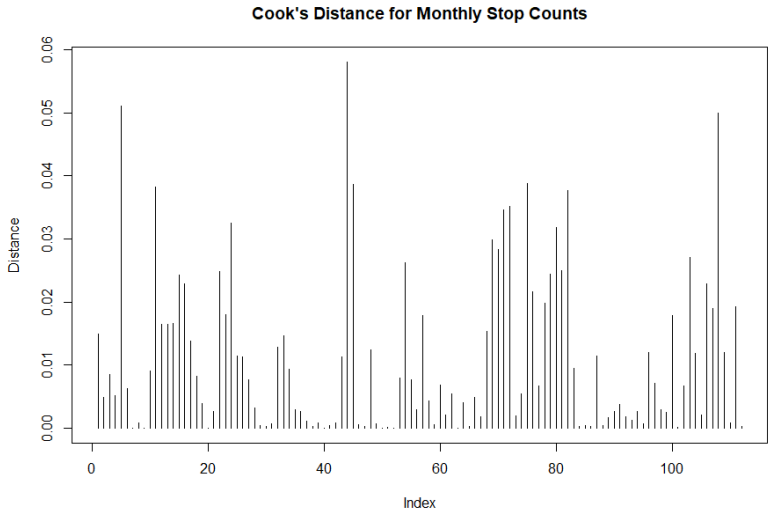


Figure 3.2 Testing of ARIMA Model Trend, Residuals, and Normality

There doesn't appear to be a notable trend, outliers, or change in variance over time in the standardized residual scatterplot below. The QQ plot indicates normality as the residuals remain close to a straight diagonal line. None of the lags in the ACF of residuals are significant above or below the autocorrelation threshold lines. Last, none of the residuals are significant in the Ljung-Box plot below the p-value threshold. Therefore, the ARIMA model is suitable for the data.

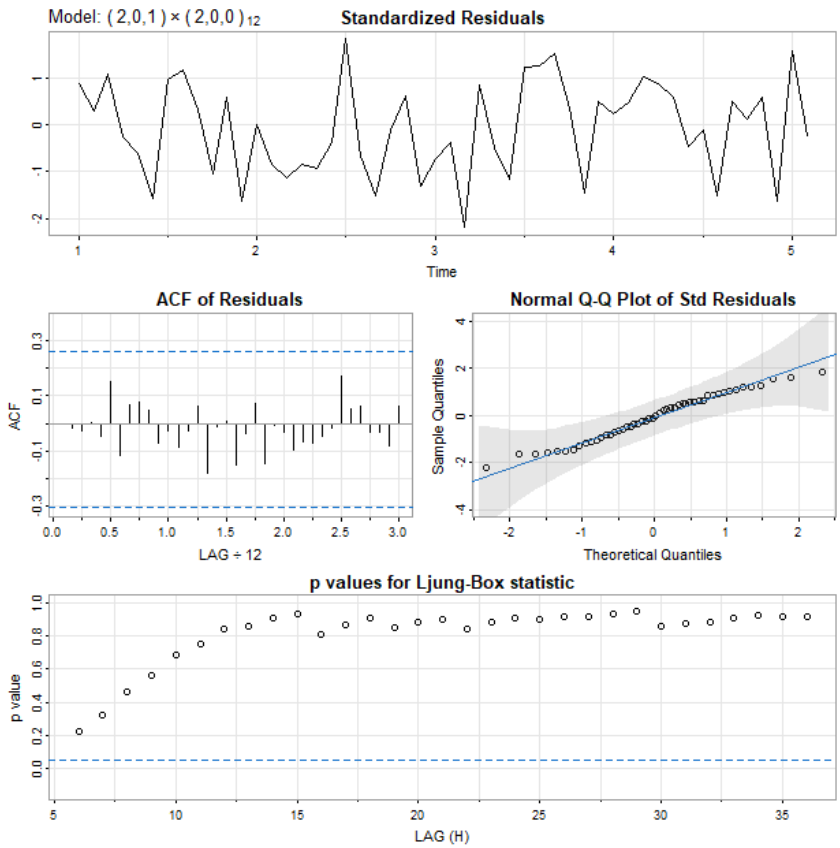
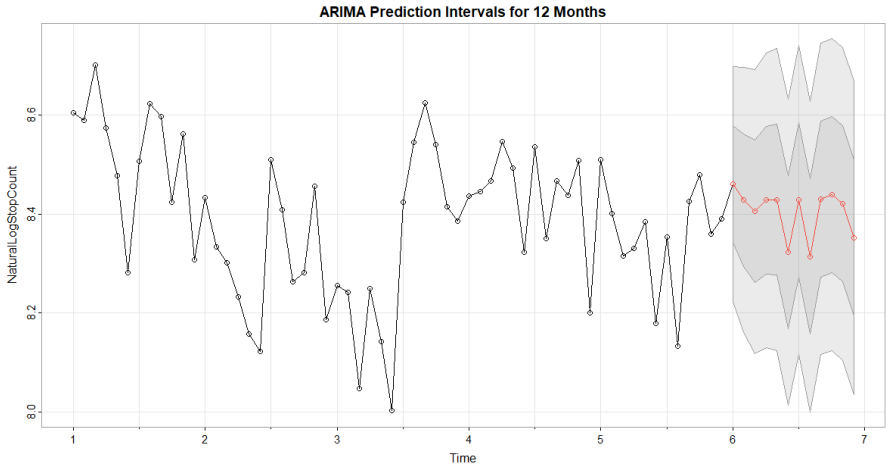


Figure 3.5 ARIMA Prediction Intervals for 12 Months After Test Set



7.3 – REFERENCES

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